|INCH-POUND|
MIL-G-80268A
25 August 1989
SUPERSEDING
MIL-G-80268
22 December 1986

MILITARY SPECIFICATION

GRINDING MACHINES, TOOL AND CUTTER; AND GRINDING AND GASHING MACHINES, TOOL, CUTTER, FLUTE, AND RADIUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, electric motor driven, manual and power operated tool and cutter grinders and tool, cutter, flute, and radius grinders with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Governments minimum requirements for each procurement.

1.2 Classification. The grinding machine shall be of the following types, styles, and sizes. The types, styles, and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Floor mounted, horizontal spindle, tilting and swiveling wheelhead, tool and cutter grinder.

Style A - Manual (operator control).

Size	Swing over table (diameter)(inches)	Maximum distance between tailstock centers (inches)
1	8	15, 22*, 24*, 32*
2	10	17*, 22, 25*, 27*, 33*
3	12	27, 32*

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 3415
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

Style B - Power operated

Size	Swing over table (diameter)(inches)	Maximum distance between tailstock centers (inches)
1	8	15, 17*, 22*, 24*, 32*
2	10	22, 25*, 27*, 33*
3	12	21* 27, 32*

Type II - Floor mounted, vertical spindle tool and cutter grinder.

Style A - Manual (operator controlled).
Style B - Power operated.

Size	Swing over table (diameter)(inches)	Maximum distance between tailstock centers (inches)
1	8	15

Type III - Floor mounted, horizontal spindle, reciprocating wheelhead, manual, tool, cutter, flute, and radius grinder.

Size	Maximum workpiece diameter between centers (inches)	Maximum workpiece length between centers (inches)
1	12-3/4	7

Type IV - Floor mounted, horizontal spindle, traversing wheelhead, manual, tool, cutter, flute, and radius gashing and grinding machine.

Size	Swing over main slide (diameter)(inches)	Maximum workpiece length between centers (inches)
1	12	6

*Indicates additional sizes available.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel,Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI B7.1 - Abrasive Wheels, Safety Requirements for the use, Care, and Protection of. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY (10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Machine types and styles.

3.2.1.1 Type I. Type I machines shall be floor mounted tool and cutter grinders. Style A machines shall be manually controlled by the operator using handwheels or levers to control the table transverse and longitudinal travel. Style B machines shall have power table travel with handwheels or levers provided for manual set-up and adjustment. The wheelhead spindle shall be mounted horizontally. The workhead shall be either a manual universal workhead or a motor driven independent workhead. Type I machines shall be equipped with attachments for external and internal cylindrical grinding, gear cutter grinding, and surface grinding as specified herein. Type I machines shall sharpen a variety of cutters including face and end mill cutters, reamers, gear cutters, counterbores, and other metal cutting tools.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.1.2 Type II. Type II machines shall be floor mounted tool and cutter grinders. Style A machines shall be manually operated using handwheels or levers to control the table transverse and longitudinal travel. Style B machines shall have power table travel with handwheels or levers provided for manual set-up and adjustment. Style B machines shall have the option of operating under automatic control. The wheelhead spindle shall be vertically mounted. The workhead shall be a manually operated universal fixture on the style A machines and a power operated universal fixture on the style B machines. Type II machines shall be equipped with attachments for radius grinding and broach grinding as specified herein. Type II machines shall sharpen a variety of cutters including face and end mills, reamers, gear cutters, and other metal cutting tools.

3.2.1.3 Type III. Type III machines shall be floor mounted tool, cutter, flute, and radius grinders. The machines shall be operated with handwheels or levers. The wheelhead spindle shall be horizontally mounted. The workhead shall be either a manually operated universal workhead or a motor driven independent workhead. Type III machines shall sharpen cutting tools having radii in multiple cutting surfaces and grind all clearances on profile, straight, and angular cutters. Type III machines shall sharpen a variety of other metal cutting tools with convex or concave radii and straight and spiral flutes.

3.2.1.4 Type IV. Type IV machines shall be floor mounted tool, cutter, flute, and radius gashing and grinding machines. The machines shall be manually operated with handwheels or levers. The wheelhead shall be horizontally mounted. The workhead shall be independently motor driven with a spiral and relief grinding attachment. Type IV machines shall sharpen conventional types of cutters including end mills, reamers, counterbores, and form cutters with straight, tapered, or helical teeth and involving convex or concave radii. Type IV machines shall make new cutters from solid stock.

3.2.2 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.2.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the X and Y axes of travel to indicate longitudinal worktable and cross slide position. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0002 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of ± 10 percent shall not adversely affect the system operation.

3.2.2.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.3 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.4 Energy efficiency. The grinding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.5 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.6 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point of operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.7 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.9 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.10 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.2.11 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.3 Construction. The grinding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the grinding machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88, and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. Drive train gears shall be manufactured of a suitable ferrous material and of the proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.3.8 Machine ways. All machine ways shall be protected, regardless of the position of machine components, to exclude dust, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. The machine ways shall be hardened to not less than 40 Rockwell "C" and ground to a 30 microinch or better finish.

3.3.9 Slides. All sliding members shall be provided with means of adjustment to meet and maintain the accuracy and alignment tolerances given in table III or IV, as applicable for the machine type, style, and size specified.

3.3.10 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. An adjustable wheel guard(s) shall be furnished for the grinding wheel(s).

3.3.11 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist basically of the following components, as applicable to the type and style machine specified in 1.2. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. Each machine base shall be a one- or two-piece cabinet type casting or stress relieved weldment. The base shall be ribbed and braced to minimize deflection and vibration under maximum operating loads and to ensure accuracies required herein. When coolant reservoirs or dust collectors are housed in the base, an access door(s) or removable plate(s) shall be provided. Means shall be provided for floor mounting and leveling of the machine.

3.4.1.1 Type I base. The base on type I machines shall be constructed with proportional V-ways, flat ways, or equal means to accept either a stationary wheelhead column and a saddle to support a traversing, sliding, swiveling worktable arrangement or a saddle to support a traversing wheelhead column and a sliding, swiveling worktable arrangement.

3.4.1.2 Type II base. Type II machines shall be constructed with a pedestal integrally attached to the base to support the vertical grinding wheelhead assembly. The base shall be constructed to also mount a vertically adjustable compound worktable column.

3.4.1.3 Type III base. The base on type III machines shall be constructed with proportional flat or V-ways to accept the main wheelhead slide. The base shall be constructed to also mount a rotating compound worktable arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.1.4 Type IV base. Type IV machine bases shall be constructed with proportional flat or V-ways to accept a traversing main wheelhead slide. The base shall be constructed to also mount a rotating compound worktable arrangement.

3.4.2 Worktable arrangement. Unless otherwise specified (see 6.2), all worktable component travel and characteristics shall meet the requirements given in table I or table II, as applicable, for the machine type, style, and size specified.

3.4.2.1 Type I worktable arrangement. The saddle on type I machines shall be mounted on base ways and shall travel transversely. The saddle shall support either the wheelhead column or the sliding, swiveling worktable arrangement. a sliding table shall be mounted either directly on the base or on the saddle and shall travel longitudinally. A T-slot and adjustable dogs or a table dog rack with adjustment screws shall be provided to reverse table motion. A minimum of two dogs shall be provided. The top of the sliding table shall be machine finished to accept a swivel table. The swiveling workholding table shall be provided with graduations to indicate table swivel in degrees and inches per foot. A fine screw adjustment shall be provided for taper settings from 0 to 2 inches per foot, indicated on a scale at the end of the table. The top of the swiveling workholding table shall be constructed with a T-slot conforming to ANSI B5.1 or comparable means for attaching a workholding device. The saddle transverse feed mechanism shall be hand operated with a fine screw adjustment having a dial conforming to 3.2.2.1. When specified (see 6.2), means shall be provided for selecting fine and coarse crossfeed on all machines. Table longitudinal feed on style A machines shall be manually operated using handwheels or levers. Table longitudinal feed on style B machines shall be infinitely variable, power operated, with a manual mode of operation for set-up work, utilizing handwheels or levers. Controls shall be provided for selecting rates of power longitudinal travel on the table. Means shall be provided to lock all moving components after positioning.

3.4.2.2 Type II worktable arrangement. Type II machines shall have a vertically adjustable worktable column supported by the base. A hand wheel shall be provided to manually raise and lower the column. Covers shall be provided on the adjustable column to protect it from contaminants in any position. A swiveling, compound worktable shall be mounted on the adjustable column. The compound worktable shall swivel about a fixed center and be provided with graduations for indicating table swivel in degrees. The compound worktable shall provide longitudinal and transverse travel for machine set-up. Longitudinal and rotational worktable and saddle motion shall be manually operated by handwheels having graduated dials conforming to 3.2.2.1. Means for clamping the adjustable column and swiveling table after positioning shall be provided. The table top shall be provided with three T-slots conforming to ANSI B5.1 or comparable means for mounting an angular work support fixture used for supporting the universal work spindle or a single T-slotted, vertically swiveling table. Each machine shall have ample bearing surface to assure permanently accurate alignment. When specified (see 6.2), automatically controlled machine cycle (with handwheels available for manual machine set-up) shall be provided. The automatic cycle shall allow the machine, after set-up, to complete roughing passes, finishing or spark out passes, and stop the machine upon cycle completion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.2.3 Type III worktable arrangement. The workholding unit on type III machines shall be supported by a turntable mounting and shall swivel in the horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. The turntable mounting shall mate directly with the base. The swiveling turntable shall carry a screw-operated cross slide to allow offsetting the axis of the workhead spindle from the center of the turntable. A top slide which supports the work holding unit shall be mounted on the cross slide. The top slide shall be screw-operated and shall allow adjustment of the workholding unit in front and behind the center of the swiveling table to adjust for various workpiece lengths. Stop dogs and dials, graduated in accordance with 3.2.2.1, shall be provided to allow accurate positioning of the workholding unit slides. Means shall be provided to lock all moving components after positioning.

3.4.3. Type IV worktable arrangement. The workholding unit on type IV machines shall be supported by a turntable mounting and shall swivel in the horizontal plane. Adjustable stops shall be provided to limit swivel in both directions. The turntable mounting shall mate directly with the base. The swiveling turntable shall carry a manual screw-operated cross slide to allow offsetting the axis of the workhead from the center of the turntable. A top slide carrying the workholding unit shall be mounted on the cross slide and shall be manually screw-operated, allowing adjustment of the workholding unit in front and behind the center of the swiveling table for various workpiece lengths. The workholding unit housing shall tilt above and below a horizontal position. Micrometer stop dogs and dials, graduated in accordance with 3.2.2.1, shall be provided to allow accurate positioning of the workholding unit slides. A positive stop shall be provided to bear against the rear surface when the head is centrally located. Means shall be provided to lock all moving components after positioning.

3.4.4 Wheelhead. Unless otherwise specified (see 6.2), the wheelhead unit shall meet the travel and characteristic requirements given in table I or table II, as applicable for the machine type, style, and size specified. The wheelhead on each machine shall be of the self-contained type. The wheelhead spindle shall be of high grade steel, hardened and ground, and mounted on precision, preloaded, antifriction bearings with sealed-in lifetime lubrication or cartridge-type mounting. The spindle shall provide vibrationless operation of the machine up to its maximum rated RPM. Grinding wheel motors shall be mounted on the wheelhead assemblies and shall be provided with means to maintain belt tension unless nontensioning belts are used.

3.4.4.1 Type I wheelhead. The wheelhead on type I machines shall be mounted on either a wheelhead column mated to the base or a wheelhead saddle slide and shall be vertically adjustable using an elevating wheel graduated in accordance with 3.2.2.1. The wheelhead shall swivel in a horizontal plane not less than 106 degrees both sides of zero and shall tilt in a vertical plane not less than 15 degrees above and below center. Wheelhead swivel and tilt shall be manually set with dials or scales graduated in 1 degree increments or less to indicate the angular setting. The wheelhead spindle shall accept straight, cup, or dish wheels. Unless otherwise specified (see 6.2), two or more spindle speed changes shall provide 3,800 to 5,500 RPM using 3-1/2 to 6 inch diameter wheels. The spindle shall be designed to permit removal of the spindle assembly from the machine as a unit for servicing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.4.2 Type II wheelhead. The wheelhead on type II machines shall be mounted on a reciprocating ram and the grinding wheel spindle shall be mounted vertically. When specified (see 6.2), an optional horizontal grinding wheel spindle shall be provided. The grinding wheel shall be vertically adjustable for set-up. The wheelhead shall also be horizontally adjustable on the ram and shall tilt 90 degrees left of zero and 70 degrees right of zero. Dials or scales graduated in 1 degree increments or less shall be provided to indicate angular settings. The spindle shall accept cup wheels. Two or more spindle speed changes shall provide wheel speeds from 3,200 to 4,800 RPM using 4, 5, or 6 inch diameter wheels. When automatic cycle is provided (see 3.4.2.2), the cycle shall allow the operator to load the workpiece, establish such parameters as ram reciprocation, cutter index, wheel spindle rotation, feed and work spindle traverse, and place the machine in automatic mode to operate unattended during a machine cycle.

3.4.4.3 Type III wheelhead. On type III machines, the wheelhead unit shall consist of sliding members, permitting transverse and longitudinal adjustment. A main slide shall mate with the base and provide longitudinal adjustment. A wheelhead cross slide shall be mounted on the main slide and provide transverse adjustment of the wheelhead. The wheelhead unit shall be manually adjustable. Scales shall be provided to indicate longitudinal and transverse adjustment. The spindle shall accept straight wheels. Two or more spindle speed changes shall provide 3,800 to 5,800 RPM using 4 to 6 inch diameter wheels.

3.4.4.4 Type IV wheelhead. The wheelhead unit on type IV machines shall consist of three sliding members providing longitudinal, transverse, and vertical adjustment of the unit. A wheelhead slide shall mate with the base providing transverse movement. Another longitudinal slide and a vertical slide shall mount on this base slide to provide longitudinal movement and vertical adjustment of the wheel. The slides shall be manually operated using handwheels or handcranks graduated according to 3.2.2.1. The wheelhead unit shall be fitted with a permanently mounted clearance angle scale to provide quick setting of the grinding wheel for any desired clearance angle in both the vertical and transverse directions. The spindle shall accept straight, dish, and cup wheels. Three spindle speeds shall provide 4,000 to 8,000 RPM wheel speed using 3-1/2 and 4 inch diameter wheels.

3.4.5 Workhead. The workhead spindle on each machine shall be hardened and ground alloy steel, and shall run in antifriction bearings. Spindle tapers shall be manufacturer's standard and shall conform to ANSI B5.10.

3.4.5.1 Type I workhead. The workhead on type one machines shall be mounted on the swiveling workholding table (see 3.4.2.1). The workhead shall be accurately aligned on the table and securely held in position. Unless otherwise specified (see 6.2), the workhead shall be universal, manually operated, and shall swivel 360 degrees in a horizontal plane and tilt at least 90 degrees in a vertical plane. All angular settings shall be indicated on dials or scales graduated in increments of 1 degree or less. The workhead shall be complete with center and draw-in bolt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.5.2 Type II workhead. Unless otherwise specified (see 6.2), type II machines shall be provided with centers, an arbor, and a manually operated, universal workhead mounted on a subtable. The centers, the subtable, and the universal workhead shall mount an angular work support fixture capable of tilting 180 degrees in a vertical plane. The angular work support fixture shall mount in the worktable T-slots (see 3.4.2.2). All angular settings shall be indicated on dials or scales graduated in increments of 1 degree or less. The workhead shall be complete with center and draw-in bolt.

3.4.5.3 Type III workhead. The workhead on the type III machines shall be mounted on a vertical master slide bolted on the worktable slide of the worktable arrangement (see 3.4.2.3). The workhead shall be vertically and transversely adjustable on the vertical and worktable slides. Unless otherwise specified (see 6.2), the workhead shall be a manually operated, universal workhead capable of pivoting the axis of the cutter plus or minus 40 degrees from horizontal. The workhead shall revolve about the axis of the cutter an amount equal to the helix angle of the cutter. Machine design shall permit making a gash to center with negative, positive, or radial rake with adjustments to secure the desired rake angle. The workholder shall have means for indexing 2, 3, 4, 5, 6, and 8 fluted cutters. A tooth rest shall be provided for cutters with more than 4 flutes. A dial or scale indicating desired clearance for grinding end mill cutters shall be provided. The workholding spindle shall accept adapters for reducing collets, as specified in 3.8.4 (f).

3.4.5.4 Type IV workhead. The workhead on type IV machines shall be mounted on the main slide of the worktable arrangement with a suitable mechanism operating through a spiral lead device. A set of index plates shall be provided with equal divisions of 8, 10, and 12. A workhead spindle shall also be fitted with a permanently mounted, direct indexing disk, a spring loaded micrometer, and an adjustable index finger for ratchet indexing of 10, 12, and 14 divisions. An adjustable tooth rest shall be provided for mounting on the grinding wheel spindle housing. The spindle shall accommodate split collets. A set of index plates for a helical lead mechanism to index 8, 10, and 12 divisions shall be furnished with the workhead unit.

3.4.6 Headstock. Type I, style B machines shall be provided with an independent motor driven headstock unit, capable of being removed from the table and shall be adjustable along the length of the table. The headstock shall have a horizontal tilt adjustment, graduated in one degree increments, of not less than 90 degrees both sides of zero alignment with the table. The headstock spindle shall be hardened and ground alloy steel with a surface hardness of not less than Rockwell 48 on the "C" scale. The spindle shall be mounted in preloaded antifriction bearings, arranged for either dead or live center grinding. The spindle shall have the manufacturer's standard taper, fitted with a center and shall have provisions for mounting chucks and faceplates. A live center pulley shall be provided. The headstock shall meet the requirements in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.7 Tailstock. When specified (see 6.2), a tailstock(s) shall be provided. The tailstock shall be adjustable along the table in alignment with the headstock. The tailstock taper, or other internal configuration, shall be as specified. The spindle shall act against a spring to allow tension adjustment, and shall be provided with a clamping device for locking the spindle in position.

3.4.8 Coolant system. When specified (see 6.2), a flood type or a mist type coolant system shall be provided. The flood type coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.9 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.9.1 Hydraulic motors. Hydraulic motors shall conform to or exceed the requirements of JIC H-1.

3.4.10 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed for operation on 3-phase, 60-hertz (Hz), dual 230/460-voltage. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.10.1 Motors. Motors shall meet the requirements of 3.4.10. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (ac) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.10.2 Worklight. When specified (see 6.2), a worklight conforming to the requirements of 3.4.10 shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.4.10.3 Hour meter. When specified (see 6.2), the grinding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.11 Dust collector. When specified (see 6.2), a dust collection unit shall be provided. The dust collector shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel. The dust collector shall be either a built-in unit or a separate external unit.

3.4.12 Optical comparator. When specified (see 6.2), an optical comparator of the specified size shall be provided with the machine. The optical comparator shall allow the operator to measure tool diameters, lengths, angles, and radii without removing the tool from the tool holding device. The comparator shall be used at either the left or right-hand grinding wheel on dual wheel machines and on swiveling wheelheads at any angle on the column. The comparator shall tilt upwards when not in use. Unless otherwise specified (see 6.2), the comparator shall be provided with a cross line screen with protractor ring, tool makers chart, vernier, a 20X screen, a positioning device giving longitudinal and cross adjustment, a power supply, a halogen bulb light, and brackets for mounting the comparator on the machine. The comparator shall be mounted to allow simultaneous viewing of the workpiece and grinding wheel.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall be not less than those specified in tables I, II, or III for the machine type, style, and size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in tables IV and V when tested using the procedures given in 4.8.3.

3.7 Performance. The machine shall be capable of sharpening conventional type cutters to like new condition tolerances and finishes. The machine shall grind workpieces in accordance with the test procedures given in 4.8.1, 4.8.4.1, and 4.8.4.2. Diameters of cutter tooth edges, when measured at any point along their full length, shall be within +0.000 inch to -0.0015 inch of a desired dimension. Runout shall not exceed 0.0015 inch TIR. Surface finish shall not be rougher than 16 microinch.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. Unless otherwise specified (see 6.2), the following standard equipment shall be furnished for the machine type, style, and size specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.8.1 Type I standard equipment.

a. One grinding center.

b. Left and right-hand tailstocks with centers.

c. Set of work driving dogs.

d. All tooth rests, tooth rest bars, holders, tooth rest holder clamps, and other tooth rest equipment normally furnished with the grinder.

e. Wheel sleeves, mounts, collets, and mandrels normally furnished with the machine.

f. Center height gage.

g. Reducing collets, the number of each and sizes shall be as specified (see 6.2). Tapers shall match workhead and headstock (see 3.4.4.1 and 3.4.5).

h. One industrial grade diamond of at least 0.20 carat, complete with holder and bracket.

i. All wrenches, cranks, handles, clamps, cutter bars, wheel guards, wheel pullers, and T-bolts normally furnished with the grinder.

j. One set of grinding wheels (not less than two) capable of grinding carbon steel and high speed steel.

3.8.2 Type II standard equipment.

a. Attachment equipped with centers, fitted to the universal work support fixture.

b. Universal tooth rest, plain rest, blades, and all other tooth rests furnished commercially with the machine.

c. Centering gage.

d. Wheel mounts, mandrels, collets, and sleeves normally furnished with the machine.

e. All wrenches, cranks, handles, wheel guards, and T-bolts normally furnished with the machine.

f. One set of grinding wheels (not less than two) capable of grinding carbon steel and high speed steel.

g. One industrial diamond of at least 0.20 carat, complete with holder and bracket.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

h. Reducing collets, the number of each and sizes shall be as specified (see 6.2). Tapers shall match the universal work support fixture.

3.8.3 Type III standard equipment.

a. All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, T-bolts, and stock supports normally furnished as standard equipment with the machine.

b. Cutter tooth setting gage.

c. Universal tooth rest and holder.

d. One set of grinding wheels (not less than two) capable of grinding carbon steel and high speed steel cutters.

e. One diamond of at least 0.20 carat, complete with holder and bracket.

f. Ball bearing work holding spindle bored one of the following ways, as specified (see 6.2):

(1) For #11 B&S taper shank, with 11 to 9 and 11 to 7 reducing collets.

(2) As required by the procuring agency.

3.8.4 Type IV standard equipment.

a. All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, T-bolts, and stock supports normally furnished as standard equipment with the machine.

b. Tooth rest assembly with one extension bar.

c. Tooth rest centering gages.

d. Set of three index plates for helix lead indexing device, 8, 10, and 12 divisions.

e. Set of 13 straight cylindrical collets with adapters as required in sizes 1/8, 3/16, 1/4, 5/16, 3/8, 7/16, 1/2, 5/8, 3/4, 7/8, 1 inch, 1-1/8 and 1-1/4 inches.

f. Set of six taper collets, compatible with the workhead spindle. The size of the workholding taper shall be as specified (see 6.2).

g. One set of grinding wheels (not less than two) capable of grinding carbon steel and high speed steel.

h. One diamond of at least 0.20 carat, complete with holder and bracket.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.9.1 Type I optional equipment.

a. One cylindrical grinding attachment, complete with 3-, 4-, or 6-jaw universal chuck, two sets nonreversable chuck jaws (one set internal and set one external), chuck adapter, spindle assembly, driving motor, and all other associated components required. Chuck size, spindle taper, and motor horsepower shall be as specified (see 6.2).

b. One internal grinding attachment complete with mounting bracket, wheel spindle housing, and spindle (for use with removable quills) driven by the wheelhead spindle through a pulley and belt or a self contained motor. The number of quills, their sizes, and the number of grinding wheels, their sizes, types, dimensional characteristics, and material composition shall be as specified (see 6.2).

c. One surface grinding attachment, complete with swivel vise (360 degrees in both horizontal and vertical planes), intermediate support, base plate, and necessary T-bolts.

d. One gear cutter sharpening attachment including all necessary hardware for grinding the cutter sizes as specified (see 6.2).

e. Radius grinding attachment with radius and maximum cutter capacity as specified (see 6.2), including swivel housing and table, top plate, workhead support and locating gage.

f. Form relieving fixture. Manual or motorized and spindle collet size shall be as specified (see 6.2).

g. Grinding wheels, number of each size, type, dimensional characteristics, and material composition shall be as specified (see 6.2).

h. Other equipment as specified by the procuring activity.

3.9.2 Type II optional equipment.

a. One radius grinding fixture for cutters with a maximum diameter of 6 inches and radii up to 1-1/2 inches with a #12 B&S taper or #50 international taper as specified (see 6.2).

b. One broach grinding fixture for flat broaches up to 3-3/4 inches wide.

c. One tool vise workholding fixture.

d. Grinding wheels, number of each size, type, dimensional characteristics, and material composition shall be as specified (see 6.2).

e. Other equipment as specified by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

3.9.3 Type III optional equipment.

a. One center assembly to mount tools between centers.

b. One motor driven workhead with variable speeds. Spindle taper size shall be as specified (see 6.2).

c. Grinding wheels, number of each size, type, dimensional characteristics, and material composition shall be as specified (see 6.2).

d. Other equipment as specified by the procuring activity.

3.9.4 Type IV optional equipment.

a. One internal grinding attachment complete with 1/4-inch collet.

b. One large cutter radius grinding attachment.

c. One 4-inch universal 3-jaw chuck for universal workhead.

d. Additional index plates for spiral lead indexing unit as specified (see 6.2).

e. Grinding wheels, number of each size, type, dimensional characteristics, and material composition shall be as specified (see 6.2).

f. Other equipment as specified by the procuring activity.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or Military specification for each lubricant (if applicable)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:33Z
Check the source to verify that this is the current version before use.

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or Military or Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the grinding machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.4. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The grinder shall be examined to determine compliance with all requirements of this specification.

4.8 Tests. Unless otherwise specified (see 6.2), each machine shall be tested as specified in 4.8.1 through 4.8.4. Failure of any machine to pass any test shall be cause for rejection. All instruments, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The grinding machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance tests given in 4.8.4 through 4.8.4.2. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The performance of each grinder shall be tested as follows for the type machine specified. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Type I and type II. Grind or regrind a 4 by 1-1/2 inch, high speed steel, left- or right-hand, arbor mounted type, helical, plain milling cutter having a minimum of 10 cutting teeth. Finished cutter physical dimensions shall conform to the tolerances given in 3.7.

4.8.4.2 Type III and IV machines. Grind or regrind a 1 by 1 5/8 inch, multiple fluted, high speed steel, left- or right-hand, shank mounted type, helical end mill. Shank tapers shall conform to ANSI B5.10. Finished cutter dimensions shall conform to 3.7.

4.8.4.3 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.11 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1. Types I and II. The tool and cutter grinders covered by this specification are intended for resharpening conventional types of cutters within the range specified, including all types of milling cutters, reamers, counterbores, hobs, taps, form tools, thread chasers, or other metal cutting tools. With the proper attachments, they shall be used for internal and external cylindrical grinding and surface grinding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

6.1.2 Types III and IV. The tool, cutter, gashing, flute and radius grinders covered by this specification are intended for sharpening conventional-type end milling cutters having cylindrical or tapered sides, straight or helical flutes, and ball, radius, or square nose. With proper attachments, they shall be used for grinding and sizing shell or ball end mills, face-milling cutters, and form-milling cutters that can be mounted on arbors within the capacity of the machine. Type IV machines shall also be used to make new cutters from solid stock.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.2).

f. Digital readout system, if required (see 3.2.2.2).

g. Specify any special digital readout system features, if required (see 3.2.2.2).

h. Additional safety and health requirements, if required (see 3.2.6).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.6).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.11).

l. Worktable arrangement, if different (see 3.4.2).

m. Fine and coarse crossfeed, if required (see 3.4.2.1).

n. Automatic cycle on type II machines, if required (see 3.4.2.2).

o. Wheelhead, if different (see 3.4.4).

p. Wheelhead speed on type I machines, if different(see 3.4.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

q. Horizontal grinding wheel spindle on type II machines, if required (see 3.4.4.2).

r. Workhead on type I machines, if different (see 3.4.5.1).

s. Workhead on type II machines, if different (see 3.4.5.2).

t. Workhead on type III machines, if different (see 3.4.5.3).

u. Tailstock, if required; specify taper or internal configuration (see 3.4.7).

v. Coolant system, if required; specify type (see 3.4.8).

w. Electrical system, if different (see 3.4.10)

x. Initial voltage, if different (see 3.4.10).

y. Motors, if different (see 3.4.10.1).

z. Worklight, if required (see 3.4.10.2).

aa. Hour meter, if required (see 3.4.10.3).

bb. Dust collector, if required (see 3.4.11).

cc. Optical comparator,if required; specify size (see 3.4.12).

dd. Features of optical comparator, if different (see 3.4.12).

ee. Size and capacity, if different (see 3.5).

ff. Additional performance features, if required (see 3.7.1).

gg. Standard equipment, if different (see 3.8).

hh. Reducing collets, as required (see 3.8.1 (g) and 3.8.2 (h)).

ii. Ball bearing workholding spindle, bored as required (see 3.8.3 (f)).

jj. Workholding taper size, as required (see 3.8.4 (f)).

kk. Specify and fully describe optional equipment, if required (see 3.9).

ll. Chuck size, spindle taper, and motor horsepower required for cylindrical grinding attachment for type I machines (see 3.9.1 (a)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

mm. Number of quill sizes and number of grinding wheels, their shank size, type, dimensional characteristics, and material composition required for internal grinding attachment for type I machines (see 3.9.1 (b)).

nn. Cutter sizes required for gear cutter sharpening attachment for type I machines (see 3.9.1 (d)).

oo. Specify radius and maximum cutter capacity required (see 3.9.1 (e)).

pp. Form relieving fixture , if required, specify manual or motorized and spindle collet size (see 3.9.1 (f)).

qq. Size, number, type, dimensional characteristics, and material composition of grinding wheels (see 3.9.1 (g)).

rr. Radius grinding fixture taper configuration, as required (see 3.9.2 (a)).

ss. Size, number, type, dimensional characteristics, and material composition of grinding wheels (see 3.9.2 (d)).

tt. Motor driven workhead spindle taper size, as required (see 3.9.3 (b)).

uu. Size, number,type, dimensional characteristics, and material composition of grinding wheels (see 3.9.3 (c)).

vv. Size of additional indexing plates, if required (see 3.9.4 (d)).

ww. Size, number, type, dimensional characteristics, and material composition of grinding wheels (see 3.9.4 (e)).

xx. Specify applicable repair parts, if required (see 3.10).

yy. Lubrication plate, if required (see 3.11.1).

zz. Information to be provided on lubrication plate, if different (see 3.11.1).

aaa. Nameplate, if different (see 3.11.2).

bbb. Hydraulic plate, if different (see 3.11.3).

ccc. Information to be provided on hydraulic plate, if different, (see 3.11.3).

ddd. Inspection conditions, if different (see 4.3).

eee. First article inspection, if different (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

fff. Quality conformance inspection, if different (see 4.5).

ggg. Acceptance test, if different (see 4.6).

hhh. Test requirements, if different (see 4.8).

iii. Additional performance test, if required (see 4.8.4.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD From 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first 10 production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.8.1 and 4.8.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

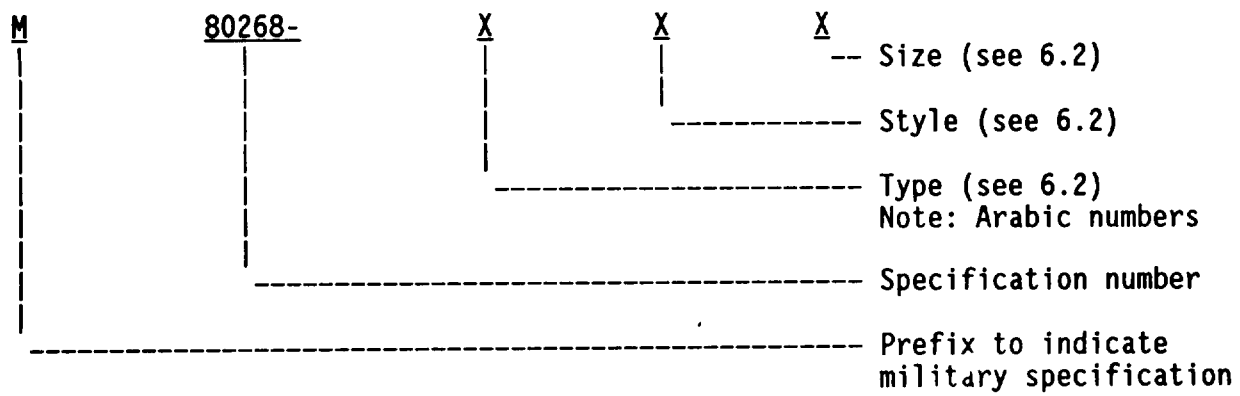
6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

6.14 Subject term (keyword) listing.

Abrasive, Wheel
Cutting
Equipment, Industrial, Plant
Machine, Shop
Manufacturing
Mounting, Floor
Operation, Power, Manual
Sharpening

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities type I machines

Characteristics	Style A			Style B		
	Size 1	Size 2	Size 3	Size 1	Size 2	Size 3
Capacity:						
Swing over table, inches	8	10	12	8	10	12
Distance between centers, inches						
of tailstock	15	22	27	15	22	27
of workhead and tailstock	12	17-1/2	24	11	20	24
of headstock and tailstock	----	----	----	11	20	24
Distance between centerline of wheel spindle to table top, minimum to maximum, inches	5-3/8 to 9-1/2	5-3/8 TO 10-1/2	5-3/8 TO 13-1/2	3-3/4 TO 10-3/4	3-1/4 TO 10-1/2	5-3/8 TO 13-1/2
Sliding table arrangement:						
Work surface, inches	19 x 4	36 x 5	44 x 6	24 x 4	30 x 5	44 x 6
Longitudinal travel, inches	10	14-1/2	24	10	14-1/2	24
Power or manual table movement	<----Manual---->			<----Power or Manual---->		
Power table movement						
Number of feeds	----	----	----	6	7	infinite
Feedrate, ipm	----	----	----	7 to 100	28 to 90	2 to 200
Cross travel, inches	4-1/2	7-/34	8	7	8	8
Horizontal swivel, each side zero, degrees	<----90---->			<----90---->		
Wheelhead or grinding unit:						
Number of spindle speeds	2	2	4	3	2	4
Wheel speed range, rpm	<----3,800 to 5,500---->					
Vertical adjustment of wheelhead, inches	6	7	10-1/2	8	7-1/4	10-1/2
Wheelhead horizontal swivel, degrees each side of zero	<----110---->			110	106	110
Wheelhead vertical tilt, degrees, each side of zero	<----15---->			<----15---->		
Motor horsepower	1/2	3/4	1	<----1---->		
Workhead or headstock:						
Manual or power	<----Manual---->			<----Manual or Power---->		
Number of speeds	----	----	----	128 to 540	128 to 400	265 to 370
Horizontal tilt, degrees, each side of zero	<----90---->			<----90---->		
Motor horsepower	----	----	----	1/6	1/4	1/4

NOTE 1: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2, nor shall they be greater than the next larger size shown in table I above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

NOTE 2: Dashes (----) indicate feature not available on machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

TABLE II. Sizes and capacities for type II machines.

Characteristic	Type II, Styles A and B Size 1
Capacity:	
Swing over table, inches	8
Maximum distance between centers, inches	15
Maximum distance between grinding wheel and base table, inches	16-3/4
Maximum cutter diameter, inches	16
Worktable:	
Carriage cross travel, inches	14
Longitudinal table travel, inches	12-1/2
Angular adjustment of carriage, degrees	180
Angular adjustment of subtable in vertical plane, degrees	45
Vertical adjustment of base table, inches	12-1/2
Center table work surface, inches	3-1/2 x 24
Base table work surface, inches	16 x 16
Work holder angular adjustment, inches	180
Wheelhead:	
Spindle speed range, rpm	3,200 to 4,800
Vertical adjustment of grinding wheel, inches	1 to 1-1/2
Angular adjustment of spindle head,	
degrees left	90
degrees right	70
Maximum horizontal travel of grinding unit, inches	7
Motor horsepower	1

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2, nor shall they be greater than the next larger size shown in table II above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

TABLE III. Sizes and capacities for types III and IV machines.

Characteristic	Type III Size 1	Type IV Size 1
Capacity:		
Maximum swing (diameter), inches	12-3/4	12
Maximum tool length, inches	7	6
Maximum radius, inches		
convex	5-1/2	5
concave	5-1/26	5
Minimum lead generated by spiral lead mechanism, inches	---	1-13/16
Workhead:		
Movement normal to cutter axis, total, inches	8-5/8	5-1/4
Movement parallel to cutter axis, total, inches	10-1/8	5-3/4
Turntable base swivel, degrees	210	235
Workhead tilt, degrees	+40	+40
Workhead vertical adjustment, inches	3	---
Wheelhead:		
Longitudinal slide adjustment, inches	7	6
Cross slide adjustment, inches	12-1/2	7-1/2
Vertical slide adjustment, inches	---	9-1/2
Grinding wheel diameter, maximum, inches	6	4
Spindle speeds:		
Number of wheelhead speeds	2	3
Wheelhead, rpm	3,800 to 5,800	4,000 to 8,000
Workhead, rpm	---	436
Motors:		
Wheelhead drive, horsepower	1	1/2
Workhead drive, horsepower	---	1/4

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2, nor shall they be greater than the next larger size shown in table III above. Provided the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

TABLE IV. Alignment tolerances for types I and II machines.

Test to be applied (as applicable to type of machine)	Maximum allowable deviation
Worktable:	
Table parallel with wheel spindle	0.002 inch per ft.
Traverse movement of table parallel with table surface	0.0008 inch per ft.
Rise and fall of table in its longitudinal travel	0.0008 inch per ft.
Universal head and tailstock:	
Workhead spindle runout, 4 inches from nose	0.0003 inch TIR
Tailstock spindle parallel with table	0.0008 inch per ft.
Tailstock spindle parallel with center t-slot of table	0.0008 inch per ft.
Tailstock axis in line with workhead axis (workhead previously set parallel with table and t-slots by means of mandrel)	
- in vertical plane	0.0004 inch TIR
- in horizontal plane	0.0004 inch TIR
Wheel spindle:	
Wheel spindle for axial slip and runout	0.0003 inch TIR
Vertical adjustment of wheelhead square with worktable:	
- in plane with long axis of table (for machines with vertically adjustable wheelhead)	0.0036 inch per ft.
- in plane perpendicular to worktable plane (for machines with vertically adjustable wheelhead)	0.0036 inch per ft.
Wheelhead swivel parallel with table surface	0.002 inch TIR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

TABLE V. Alignment tolerances for types III and IV machines.

Test to be applied (as applicable to type of machine)	Maximum allowable deviation
Swivel worktable assembly:	
Worktable top parallel with its movement toward and away from the swivel center through full length of travel	0.001 inch TIR
Worktable top parallel with its cross (transverse) movement - in 1-1/4 inch travel (5/8 inch each side of swivel center)	0.001 inch TIR
Squarness of vertical movement of the workhead with the top of the worktable in both directions (full length of travel) (type III machines)	0.002 inch TIR
Wheel spindle:	
Squareness of vertical movement of the wheel spindle with the top of the worktable in both directions (full length of travel) (type IV machines)	0.002 inch TIR
Parallelism of the cross (transverse) movement of the wheel spindle with the top of the worktable, in 4 inches	0.001 inch TIR
Parallelism of the horizontal (axial) movement of the wheel spindle with the top of the worktable, in 4 inches	0.001 inch TIR
Concentricity of the wheel spindle:	
Straight diameter	0.002 inch TIR
Face	0.001 inch TIR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - ME
Air Force - 84

User activities:
Army - SM, MU, AR
Navy - MC

Preparing activity:
DLA - IP

Project (3415-0175)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:33Z
Check the source to verify that this is the current version before use.

☆U.S. GOVERNMENT PRINTING OFFICE: 1990 - 704-834/30876

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: **In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.**

NOTE: **This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.**

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Headquarters
Defense Industrial Plant Equipment Center
Memphis, Tennessee 38114-5051

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSM
Memphis, Tennessee 38114-5051

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80268A	Grinding Machines, Tool and Cutter; and Grinding and Gashing Machines, Tool, Cutter, Flute, and Radius

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	d. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:33Z
Check the source to verify that this is the current version before use.